Course Code: PHY2702	**Course Name:** Micro and Nano Systems	
Credits: 3 (3-0-0)	**Contact hours per week:** 3 hours Theory per week	
Batch: 2023, 6th Sem **Academic Year:** 2025-26	**Semester Duration:** January – May 2026	
Instructor-in-charge: Prof. Tabish Rasheed Email - tabish.rasheed@bmu.edu.in Phone - 7082768675 Office – 10, Fourth Floor, E2 Building	**Course Coordinator –** Prof. Tabish Rasheed Email - tabish.rasheed@bmu.edu.in Phone - 7082768675 Office – 10, Fourth Floor, E2 Building	

Aim of the course: To provide introduction to the fundamental concepts related to Micro and Nano systems and to reinforce the student's knowledge of basic quantum mechanics and material science which helps in understanding the basic underlying principle of working as well as the construction of these devices.

Course Overview and Context: This is an open elective course for all the branches of Bachelor of Technology (B.Tech.) that presents the basics of the production and working of Micro and Nano systems. The course aims to provide introduction to the fundamental concepts related to Micro and Nano systems and reinforce the student's knowledge of basic quantum mechanics and material science. It helps in building an understanding of the basic underlying principle of working as well as the construction of these devices. Micro and nano fabrication are one of the latest manufacturing techniques, which are being used for all modern miniaturized systems. Examples are computer chips and integrated sensors for monitoring our environment, cars, mobile phones, medical devices and more. Micro and nano systems are set to revolutionize the world due to their varied applications and versatility.

e Outcomes: At the end of the course, students should be able to

	Course Outcome Statement	**apped Component**	**ssociated Activities / Assessment**
1	and the notions of micro/nano dimensions, Physics and Quantum Mechanics.	Skill Development	lab assignments, End Term Exam.
2	ne advanced concepts of Physics, Quantum Mechanics and fabrication techniques of micro and nano systems.	Skill Development, Employability, Entrepreneurship	Term paper, lab assignments, End Term Exam.

3	the principles of Physics and Electronics to comprehend the working and construction of micro and nano systems.	ability, Entrepreneurship	assignment, Term paper.

e Competencies: (Course Outcomes further elaborated) and Instruction schedule:

Competency	Sessions	CO
Let's comprehend the micro and nano world.	2	CO1
Introduction to Micro and Nano systems. Sensors, actuators and transducers. MEMS & NEMS. Everyday life examples of Micro and Nano Systems.	3	CO1
Why to learn Quantum Mechanics?	1	CO1
Wave, probability, wave particle duality, basic postulates of quantum mechanics.	3	CO1
Apply the concept of particle in a box to understand the principle of quantum dot, quantum wire and quantum well.	2	CO1
Apply the concept of quantum tunneling to understand the working of Scanning tunneling microscope.	2	CO2
Understand the types of solids based upon arrangement of atoms- mono-crystalline, poly-crystalline, nano-crystalline and amorphous (or non-crystalline solid). Concept of grain and grain boundaries.	1	CO2
Understand space lattice, basis and crystal structure. Unit cell and its types. Atomic packing factor (APF) and number of atoms per unit cell.	2	CO2
Understand Miller Indices. Define the methodology for determining the Miller Index for a crystal plane.	2	CO2
Understand the concept of microfabrication/nanofabrication and clean-room facility. Clean-rooms (construction and working) and types of clean-rooms.	3	CO2
Understand the working principles of bulk micromachining, surface micromachining, wafer bonding, deep reactive ion etching, LIGA (LIthographie Galvanoformung Adformung) and hot embossing.	8	CO2
Understand the working steps of photolithography and electron-beam lithography.	3	CO2
Thin films and thick films. Vapour deposition techniques-CVD & PVD. Application of vapour deposition techniques in device fabrication.	3	CO2
Understand the construction and working of capacitive type MEMS accelerometer. Applications of MEMS accelerometer.	3	CO3
Understand the construction and working of MEMS gyroscope. Applications of MEMS gyroscope.	3	CO3
Understand the construction and working of MEMS magnetic field sensor. Applications of MEMS magnetic field sensor.	1	CO3
Understand the construction and working of Nanoelectromechanical relay.	1	CO3
Understand the construction and working of nanorods based gas sensor.	1	CO3
Understand Nanotechnology. Why is nanotechnology important?	1	CO3

What is Nanoelectronics? Applications and fundamental concepts. Present status and future scope of nanoelectronics.	3	CO3

Total Hours for Course: 48

) Mapping

PO and PSO →	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3	PSO 4
↓ CO Mapping Course Mapping →																
CO1	1			2	1				1	1		1				
CO2	3	2	1	2	2				1	1		2				
CO3	3	3	1	1	2				1	1		1				

1- Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation

For details about PO, PSO, and CO-PO mapping, refer to the OBE document for B. Tech. – CSE, ME & EComE

Wise Topics

Week 1-2: Imagination of micro and nano dimensions. Introduction to design of Micro and Nano Systems. Sensors, actuators and transducers. MEMS & NEMS. Everyday life examples of Micro and Nano Systems. Why to learn Quantum Mechanics? Wave, probability, wave particle duality. Basic postulates of quantum mechanics.

Week 3-4: Particle in a box: quantum dot, quantum wire and quantum well. Quantum mechanical tunneling: Scanning tunneling microscope. Types of solids based upon arrangement of atoms- mono-crystalline, poly-crystalline, nano-crystalline and amorphous (or non-crystalline solid). Concept of grain and grain boundaries. Unit cell and its types. Atomic packing factor (APF) and number of atoms per unit cell.

Week 5-6: Miller Indices. Define the methodology for determining the Miller Index for a crystal plane. Concept of microfabrication/nanofabrication and clean-room facility. Clean-rooms (construction and working) and types of clean-rooms.

Week 7-8: Working principles of bulk micromachining, surface micromachining, wafer bonding. Working principles of deep reactive ion etching, LIGA (LIthographie Galvanoformung Adformung) and hot embossing.

Week 9: Mid Semester Exam Week

Week 10-11: Working steps of photolithography and electron-beam lithography. Thin films and thick films. Vapour deposition techniques-CVD & PVD. Application of vapour deposition techniques in device fabrication. Construction and working of capacitive type MEMS accelerometer. Applications of MEMS accelerometer.

Week 12-13: Construction and working of MEMS gyroscope. Applications of MEMS gyroscope. Construction and working of MEMS magnetic field sensor. Applications of MEMS magnetic field sensor.

Week 14-15: Construction and working of Nanoelectromechanical relay. Construction and working of nanorods based gas sensor. What is Nanoelectronics? Applications and fundamental concepts. Present status and future scope of nanoelectronics.

Week 16: Revision week

ng Resources:

Textbook:

1. Chang Liu, "Foundations of MEMS", Pearson Education Inc., 2006.

Reference Books:

1. Stephen D Senturia, "Microsystem Design", Springer International Edition, 2006.
2. Tai Ran Hsu, "MEMS and Micro systems Design and Manufacture" Tata McGraw Hill, New Delhi, 2006.
3. M.H.Bao, "Micromechanical Transducers: Pressure sensors, Accelerometers and Gyroscopes", Elsevier, Newyork, 2000.
4. Marc Madou, "Fundamentals of Microfabrication", CRC press, 2002.

E-Learning (and other) Resources: Other relevant study material, MOOCs (and other e-learning material) will be suggested and discussed during the lecture session(s) as and when deemed suitable.

sment Pattern:

nal grade will be based on the marks/ grades obtained in the end-semester exam and other assessments defined in the assessment table below. The relative grading method described in the university's academic regulations will be followed to grade the students.

Evaluation Component	Weightage (%)	Evaluation Schedule	Rubrics/Remarks
Quiz	20%	16 March 2026	● Understanding of Concepts (70%) ● Problem Solving (30%)
Assignment	20%	19 April 2026	Understanding of Concepts (40%) ● Correct explanations of concepts ● Proper use of terminology Writing skills (40%) ● Ability to write in clear and original words ● Logical approach towards work assigned Neat figures and formatting (20%) ● All formatting and figures must be nicely done
Term paper	20%	12 April 2026	Report (50%) ● Demonstrates understanding of entire course material ● Ability to apply concepts related to Micro and Nano Systems Presentation (50%) ● Ability to present the work done

			● Understand the basic concepts and answer questions related to the presentation
End Semester Examination (Conducted by CoE)	40%	As per University Calendar	Comprehensive Knowledge (70%) ● Demonstrates understanding of entire course material Problem Solving (30%) ● Ability to apply concepts to solve complex problems

Experiential Learning: The term paper internal assessment component will involve a review of latest research paper. The component involves the use of concepts learnt in the course for understanding the underlying principle in given research paper. A research report followed by group presentation shall firmly embed the basic principles in the mind of the student. Also, students will be engaging in lab assignment work involving micro and nano fabrication techniques.

Student Responsibilities:

- Regular and attentive during lectures/tutorial classes
- Write class notes and submit assignments on time.
- Check announcements at Maitri / Google Classroom / E-mail regularly.
- Regularly check marks and attendance on the Maitri/Google Classroom and inform if there are any concerns.

Support for Slow Learners:

- Extra classes as per the requirements.
- Additional assignments.
- Group them with advanced learners for project development.

dance Policy: As per the University policy.

Assignments: One quiz, one assignment and one term paper.

tion procedures for tests, assignments, and lab: As per the weightage mentioned above, continuous evaluation shall be undertaken.

ssignment submission policy: Student loses 50% marks for late submission except for an exceptional situation like a medical problem. The medical issue must be communicated by email with proper documentation in advance (prior to last date of submission). The last date for assignment submission shall be provided in Google classroom and that shall be used to determine the date for submission. No assignments shall be considered after 5 May 2026.

up examination/work policy: As per the University policy

iour expectations: Attend classes regularly and be punctual. The use of mobile phones during lectures and lab work is strictly prohibited. Ensure academic integrity – no copying, no cheating, focus on learning and gaining knowledge.

mic dishonesty/cheating/plagiarism: Any student violating expected behaviour will be punished according to the University's rules and regulations.